

DESIGN, DEVELOPMENT AND TEST ON REAL DATA OF AN FM BASED PROTOTYPICAL PASSIVE RADAR

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ABSTRACT

This paper describes the design, development and test of a passive covert radar (PCR) exploiting a single non co-operative frequency modulated (FM) commercial radio station as its transmitter of opportunity. The system has been conceived in order to detect low flying and low Radar Cross Section (RCS) targets even in disturbed and hostile electromagnetic environments. Preliminary measurements of the electromagnetic field intensity at the installation and test site of the prototypical PCR, suggested developing: (i) log-periodic antennas with adequate front-to-back ratio; (ii) brand new receivers characterized by state-of-art technology; (iii) adaptive signal processing; (iv) data processing based on particle filtering. The performance of the real-time system, whose refresh rate is less than five seconds, has been assessed comparing its output to truth data provided by air traffic control systems.

1. INTRODUCTION

This work aims at describing the design, development and test of a passive covert radar (PCR) exploiting a single non co-operative frequency modulated (FM) commercial radio station as its transmitter of opportunity.

The experimentation of an FM radio based bistatic radar has already been described in [2]. The main technical advances that have been achieved are:

- the use of log-periodic antennas has improved co-channel interference rejection at RF (Radio Frequency) level;
- the development of brand new receivers characterized by state-of-art technology has increased the system capability of detecting low flying and low RCS targets even in hostile and disturbed environments;
- target direction of arrival is now effectively integrated in the tracking algorithm and the application of particle filtering to confirmed tracks properly transforms target bistatic range, Doppler and bearing into Cartesian position and velocity;
- the whole processing chain may work on a single Intel® Core™2 Quad PC in less than 5 seconds (presentation console refresh rate) for each acquisition period.

This paper is organized as follows: section 2 describes the system architecture and details its hardware subsets, i.e. antennas, receivers and processing PCs. Section 3 briefly recalls signal processing algorithms and illustrates the data processing design. Section 4 describes the real-time signal/data processing and visualization chain. Real-data results are reported in section 5. Conclusions are given in section 6.

2. SYSTEM ARCHITECTURE

The PCR functional block diagram is reported in Figure 1:

- three antennas collect surveillance and reference signals;
- analog and digital receivers transform RF data into 32-bit streams;
- digital processing extracts PCR plots and tracks and sends them to the visualization console, that also displays the data provided by the Air Traffic Control (ATC) system.

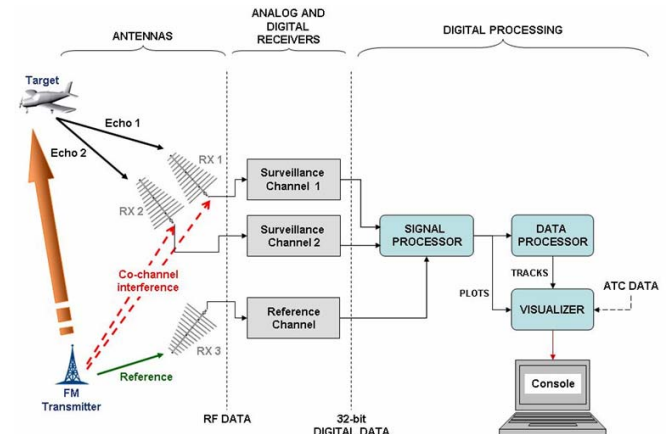


Figure 1 – PCR block diagram.

Figure 2 represents instead the system hardware architecture. The “RXA” rack contains:

1. the receivers analog section, which is connected to antennas via RF cables and is formed by the RF/IF assy and a frequency synthesizer;
2. an auxiliary section, including a fan assy, a DC/DC converter, an AC/DC converter.

The “RX ADC&DDC” rack contains:

1. the receivers digital section;

- the digital processing section, presently running on three Intel® Core™2 PCs, connected via LAN;
- an auxiliary section.

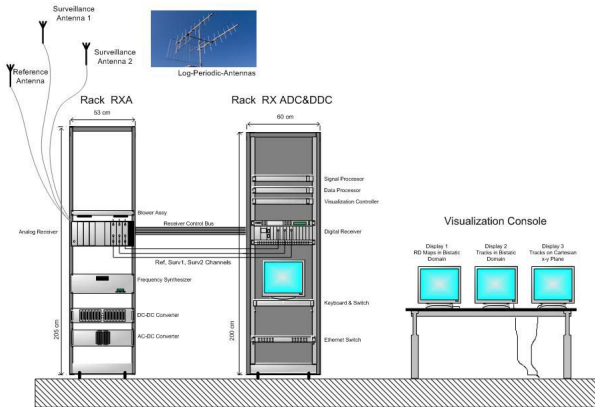


Figure 2 – PCR hardware architecture.

A picture of the PCR prototypical system installed at SELEX Sistemi Integrati site in Rome, Italy, is shown in Figure 3.

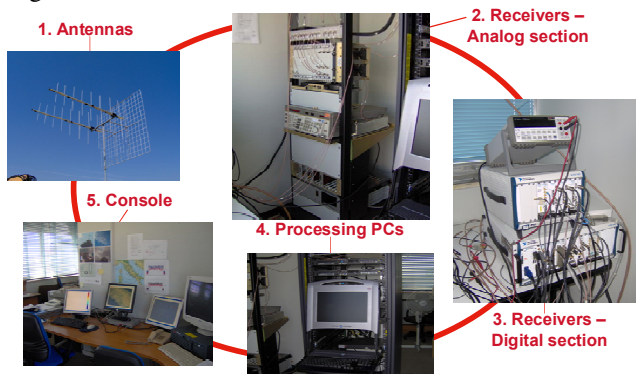


Figure 3 – PCR system at Selex Sistemi Integrati.

2.1 Antennas

Surveillance antennas have been designed in order to maximize the power ratio between target echo and co-channel interference. As for the radiation pattern, this requirement implies that the antenna gain towards the target G_{front} is maximum and the antenna gain towards the transmitter G_{back} is minimum.

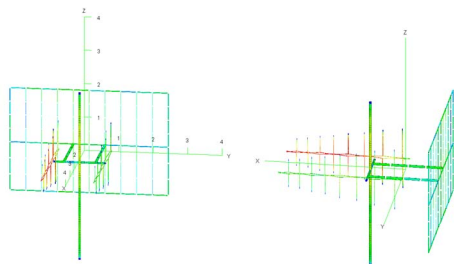


Figure 4 – Electromagnetic model of surveillance antennas.

The electromagnetic model of the surveillance structure is depicted in Figure 4. There are: (i) two log-periodic dipole array (LPDA) antennas forming an interferometer, (ii) an antenna mast, (iii) a reflecting grid. The whole structure has been simulated using the commercial software suite FEKO, based on the Method of Moments (MoM) technique. Simulation results consist of the three curves reported in Figure 5, respectively representing, from the bottom to the top, back gain, front gain and front-to-back ratio over the FM band (88÷108 MHz).

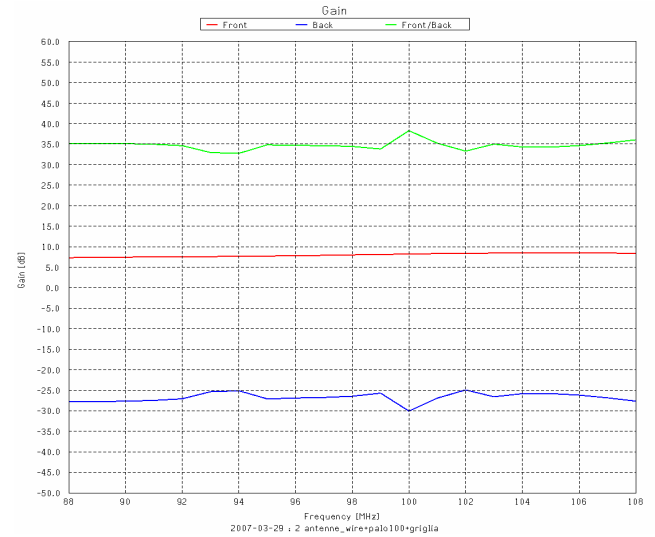


Figure 5 – Front gain (middle), back gain (lower) and front to back ratio (upper).

2.2 Receivers

Preliminary measurements of the electromagnetic field intensity at the SELEX Sistemi Integrati site in Rome, Italy, played a fundamental role in defining the requirements the PCR receivers have to comply with. In particular, they indicate:

- first, the maximum expected input power level;
- then, fixed the minimum detectable signal, the receivers dynamics.

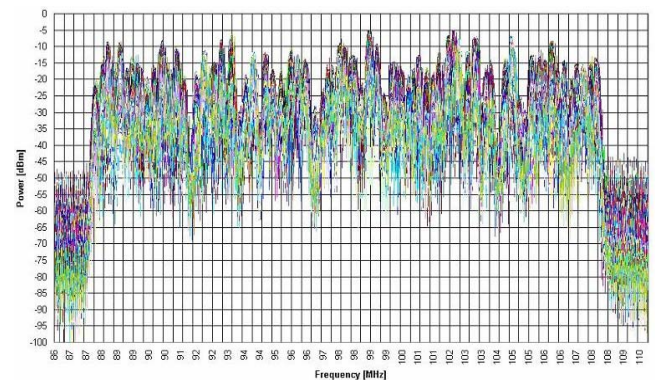


Figure 6 – FM power level measured at SELEX Sistemi Integrati site in Rome.

In the electromagnetic environment depicted in Figure 6, which is perhaps the worst condition the PCR may be expected to work in, the maximum power level the receivers have to handle is 6.3 mW (+8 dBm), while the dynamics they have to guarantee is greater than 100 dB. Both requirements are extremely demanding and have forced us to develop ad hoc channels, that make use of two conversions in the analog domain and of state-of-art technology in Analog to Digital (A/D) conversion.

The receiver block diagram is depicted in Figure 7.

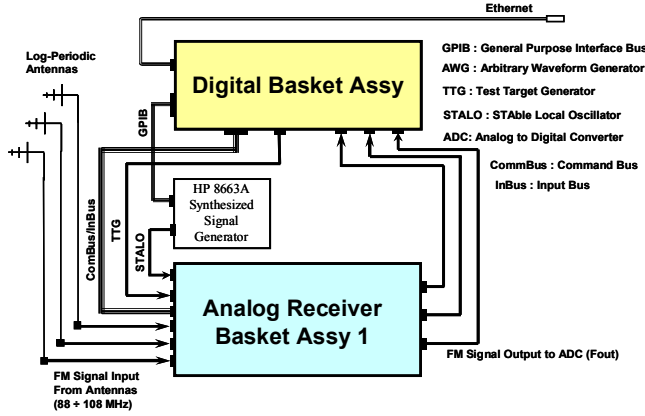


Figure 7 – Receiver block diagram.

The analog section is in charge of adjusting and transferring the RF signal from each antenna to the A/D converters. On the purpose, two down conversions are applied: the former lowers the carrier frequency from RF to IF (first IF) and, simultaneously, selects the FM channel of interest; the latter amplifies and moves the first IF signal to a lower carrier (second IF). This second conversion is necessary because the selected A/D converter, which shows a large number of bits, is characterized by a few MHz passband. The second IF signal is filtered, amplified and acquired by the A/D converter with 10 MSps clock frequency; Digital Down Conversion (DDC) is finally applied to obtain complex signal digital components handled by signal and data processor.

3. SIGNAL AND DATA PROCESSING ALGORITHMS

The signal processing block diagram is depicted in Figure 8 [2]: “Right” and “Left” correspond to the digital signals generated by the two channels of the interferometric surveillance system, whereas “Reference” is the binary data stream produced by the reference channel.

The main functions of the developed signal processor are:

- adaptive co-channel interference cancellation;
- Doppler-sensitive cross-correlation search for target echoes, using the reference signal as the optimal matched filter that provides the necessary processing gain;
- adaptive thresholding used to automatically detect targets on the two range-Doppler maps (associated to

left and right channels) maintaining a constant false alarm probability via a custom bidimensional Cell-Average CFAR (Constant False Alarm Rate);

- “2/2” logic to extract detections common to right and left channels, with ± 1 cell tolerance in Doppler;
- bearing estimation based on the use of a simple phase interferometry. The difference between the arguments of the two range-Doppler surfaces is unambiguously related to the target direction of arrival in the azimuth sector the radar surveys, which is about $\pm 30^\circ$.

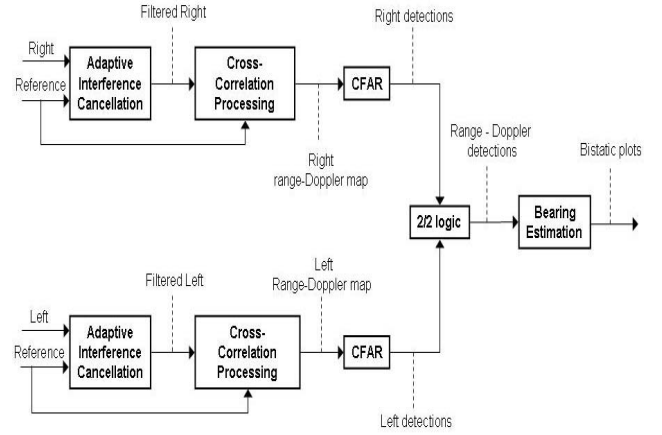


Figure 8 – Signal processing block diagram.

The block diagram in Figure 9 shows the algorithms that have been designed and developed to track the targets the bistatic passive radar detects.

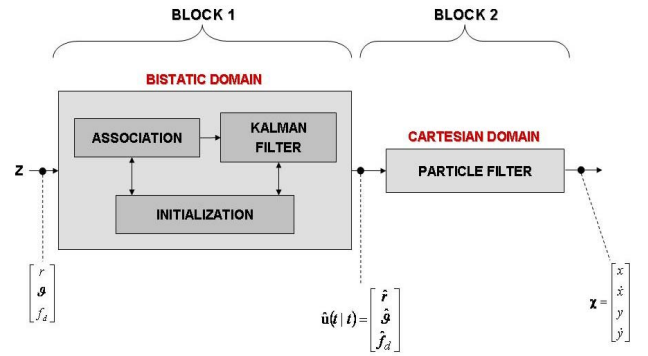


Figure 9 – Data processing block diagram.

The working principle of the implemented data processor is the following. Plots relative to slowly maneuvering targets, clutter and multipath detected in the current scan¹ are collected in the measurement vector $\mathbf{Z}$. Each plot is a set of three measurements: bistatic range r , azimuth ϑ and bistatic Doppler frequency f_d . The measurement vector $\mathbf{Z}$ feeds the tracking algorithm, which is formed by two cascading blocks. The former implements track initialization,

¹ Even if surveillance antennas do not rotate, the radar system needs about 3.5 seconds to fully process the 1 second's worth of received signals, which corresponds to a sort of conventional radars scan period.

plot to track association and recursive filtering in the bistatic domain; in particular, target motion model linearity permits using a simple Kalman filter, as suggested in [2]. The need for Cartesian tracks to be compared with simulated truth data has suggested the Kalman-filtered measurement vector should be followed by a particle filter [3]. Therefore the second block of the tracking architecture that has been implemented consists of a particle filter that applies to the output (i.e., confirmed tracks) of the previous block to estimate target tracks in the Cartesian domain. The simple geometric transformation is a low computational cost though less precise alternative to the particle filter, estimating only Cartesian target position.

Block 2 of Figure 9 receives the filtered measurement vector $\hat{\mathbf{u}}(t|t)$ from Block 1 and estimates target motion in Cartesian coordinates $\chi = [x, \dot{x}, y, \dot{y}]$, according to Equation (1):

$$\hat{\mathbf{u}}(t|t) = h(\chi) + \mathbf{v} \quad (1)$$

where

$$h(\chi) = \begin{bmatrix} r_{Rx} + r_{Tx} - d \\ \angle(x + iy) \\ -\frac{1}{\lambda} \left(\frac{x\dot{x} - y\dot{y}}{\sqrt{x^2 + y^2}} + \frac{(x - x_{Tx})\dot{x} - (y - y_{Tx})\dot{y}}{\sqrt{(x - x_{Tx})^2 + (y - y_{Tx})^2}} \right) \end{bmatrix} \quad (2)$$

The elements of the three-dimensional $h(\chi)$ in Equation (2) do respectively represent: bistatic range, bearing, bistatic Doppler frequency. They depend on:

- distance d receiver-transmitter;
- distance r_{Rx} target-receiver, and distance r_{Tx} target-transmitter;
- transmitter position $[x_{Tx}, y_{Tx}]$ (the receiver lies at the centre of the reference system).

Particles are assumed to move with uniform speed along a straight line, in accordance with Equations (3) and (4):

$$\begin{bmatrix} x(k+1) \\ \dot{x}(k+1) \\ y(k+1) \\ \dot{y}(k+1) \end{bmatrix} = \mathbf{A} \begin{bmatrix} x(k) \\ \dot{x}(k) \\ y(k) \\ \dot{y}(k) \end{bmatrix} + \mathbf{w} \quad (3)$$

$$\mathbf{A} = \begin{bmatrix} 1 & T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

where the process noise $\mathbf{w}$ has zero mean and suitable covariance matrix $\mathbf{Q}_{xy}$.

The algorithm used for particle filtering is the Sequential Importance Resampling (SIR) [1].

In our application, particle filter initialization is the major difficulty, because passive radar large measurement error makes tracking accuracy poor and because coordinates transformation from bistatic to Cartesian is ambiguous. In fact, while the correspondence between the bistatic pair $[r, \theta]$ and the Cartesian couple $[x, y]$ is univocal, unambiguity does not hold for the relation between Doppler frequency $f_d = -\frac{1}{\lambda} \frac{\partial r}{\partial t}$ and Cartesian velocity $[\dot{x}, \dot{y}]$. It

depends on the fact that Doppler shift only indicates how targets are moving with respect to the ellipse (whose foci coincide with receiver and transmitter of opportunity) they are currently lying on: if $\dot{r} > 0$, they are moving towards larger ellipses, and vice versa. Such information generates velocity uncertainty in the Cartesian domain, i.e. there are infinite couples $[\dot{x}, \dot{y}]$ generating the same Doppler frequency.

Particle filter samples are therefore initialized according to a quasi deterministic rule for position $[x, y]$ and randomly for velocity $[\dot{x}, \dot{y}]$.

$$\begin{aligned} x &= x_{init} + n_x \\ y &= y_{init} + n_y \end{aligned} \quad (5)$$

In Equation (5), n_x and n_y are Gaussian white noises with zero mean and standard deviation Δ , while x_{init} and y_{init} are evaluated using Equation (6), that transforms the bistatic estimate $[\hat{r}, \hat{\theta}]$ into the Cartesian pair $[x, y]$.

$$\begin{cases} x = \frac{(\hat{r} + d)^2 - x_{Tx}^2 - y_{Tx}^2}{2(\hat{r} + d - x_{Tx} \cos \hat{\theta} - y_{Tx} \sin \hat{\theta})} \cos \hat{\theta} \\ y = \frac{(\hat{r} + d)^2 - x_{Tx}^2 - y_{Tx}^2}{2(\hat{r} + d - x_{Tx} \cos \hat{\theta} - y_{Tx} \sin \hat{\theta})} \sin \hat{\theta} \end{cases} \quad (6)$$

The velocity vector $[\dot{x}, \dot{y}]$ is instead generated randomly, according to Equation (7).

$$\begin{aligned} \dot{x} &= V \cos(\varphi) \\ \dot{y} &= V \sin(\varphi) \end{aligned} \quad (7)$$

Absolute value V is uniformly distributed in the velocity interval of interest for PCR applications; direction angle φ is uniformly distributed in $[0, 2\pi]$.

4. REAL TIME PROCESSING AND PRESENTATION

The development of the real time system has been strictly conditioned by adaptive interference cancellation that, due

to its non parallelizable and high computational cost nature, is the processing bottleneck. In particular, two distinct programming languages have been included and combined to develop the signal processor code. The software architecture (Figure 10) is managed via C++, while a number of FORTRAN 90 subroutines specifically accomplish all signal processing functions (adaptive cancellation, cross-correlation, bi-dimensional CFAR, bearing extraction).

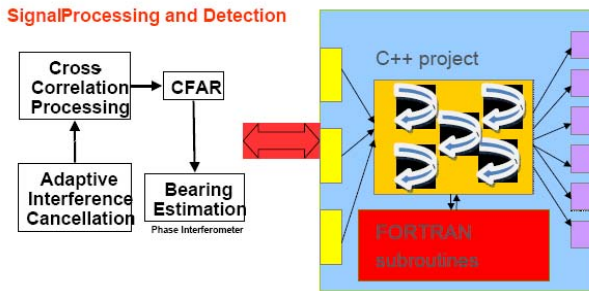


Figure 10 – Signal processing software architecture.

FORTRAN is especially suited to numeric computation and our test bed current performance rests on the fact that its use has reduced adaptive cancellation runtime from about 13 seconds in the “only C++” version to about 900 milliseconds in the “C++-FORTRAN” version. The internal “Elaboration” of the acquired data is exploded in the main calculative blocks written in FORTRAN and depicted with a red background in Figure 11.

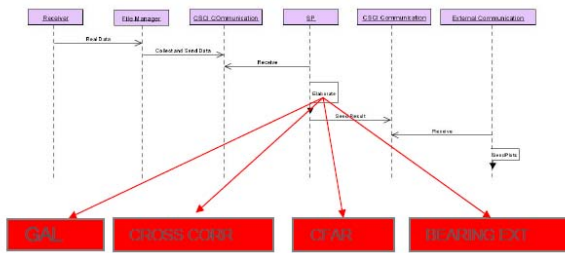


Figure 11 – Signal processing sequence diagram.

The software realization of the data processing was conditioned by the use of the particle filter and this specific part was realized using a FORTRAN subroutine (Figure 12).

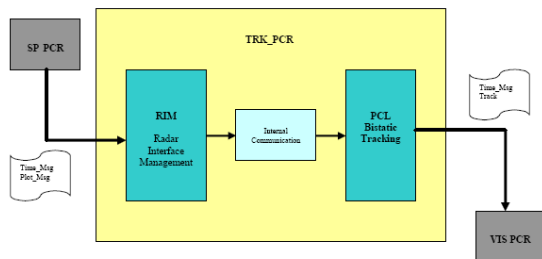


Figure 12 – Data processing software architecture.

The dedicated PPI (Plan Position Indicator) application is a three screen visualizer that allows displaying simultaneously:

1. range-Doppler maps and CFAR detections;
2. bistatic range-Doppler tracks;
3. Cartesian tracks.

The presentation console works using many operational threads in order to simultaneously show either the range-Doppler maps and plots sent by signal processor either the tracks (from data processor) onto the range-Doppler and Cartesian displays. The detail of the track data screens is presented in Figure 13.

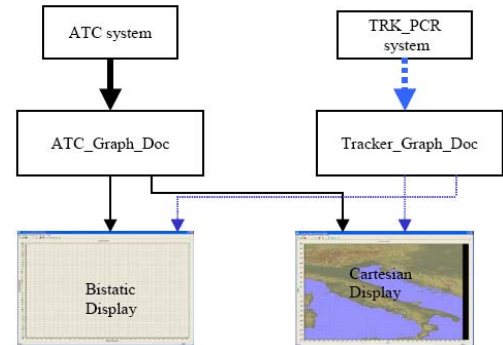


Figure 13 – Presentation console track screens architecture.

The two data flows from PCR and ATC systems are implemented using four main threads, in order to plot interlaced PCR and ATC tracks and to compare them on both track screens. The real time system runs on a Windows XP® Operative System using a commercial Intel® core™ 2 quad PC or, alternatively, employing three Intel® dual core PCs connected via LAN.

5. REAL DATA RESULTS

Some real data results are reported in the following. Figure 14 shows the three screen presentation console captured while detecting and tracking a certain number of targets, up to a bistatic range of about 360 km. On the left, the raw range-Doppler map associated to one of the surveillance channels enlightens through yellow squares the targets revealed after CFAR application. In the middle, PCR Cartesian tracks are overlapped to the geographical map they refer to; the position of the PCR receiver and of the transmitter of opportunity is labeled with RX and TX respectively. On the right, PCR tracks are depicted in the bistatic range-Doppler domain.

Performance evaluation of the PCR system has been achieved by measuring detection probability and tracking accuracy provided by the FM based passive radar with respect to ATC truth data. Figure 15 shows PCR and ATC tracks in the bistatic domain. Those ATC tracks that do not exhibit a PCR counterpart refer to angular sectors not included in the azimuth coverage of the passive radar.

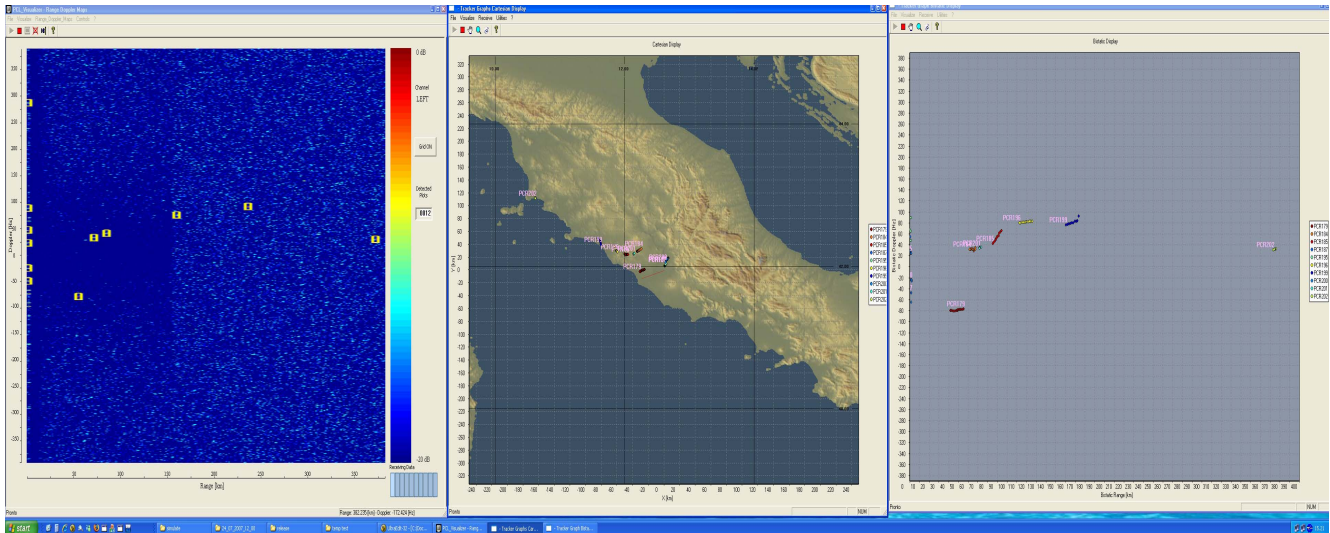


Figure 14 – Presentation console showing real data.

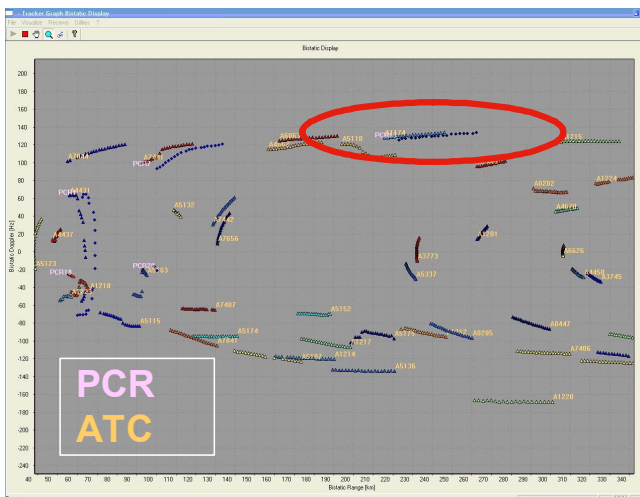


Figure 15 – Comparison between PCR and ATC tracks.

The zoom of the enlightened portion of Figure 15 is represented in Figure 16: the PCR track is initiated at about 285 km and shows a very good match with respect to the correspondent ATC track.

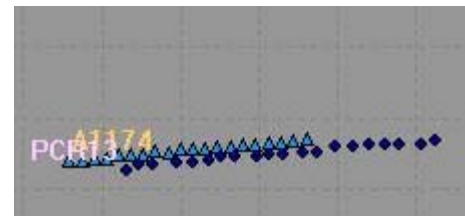


Figure 16 – Comparison between PCR and ATC tracks- zoom.

The tracking algorithm, which includes the application of particle filtering to confirmed bistatic tracks, is a noticeable step forward in the development of passive covert radars. Moreover, the comparison with ATC tracks has proved an outstanding tracking accuracy in the bistatic domain.

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6. CONCLUSIONS AND FOLLOW ON

A real time prototypical passive covert radar exploiting a single non co-operative FM radio transmitter has been designed, developed and tested. The environment the radar is installed in has forced realizing brand new antennas and receivers, in order to reject the intense co-channel interference and permit revealing low flying and low RCS targets. The signal and data processing algorithms have all been designed and developed according to radar coverage and real time requirements. This FM-band passive covert radar observes aircraft at bistatic ranges of up to 360 km in a 60° wide azimuth sector and refreshes target tracks in less than 5 seconds.